

Multimedia

Vorlesung

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Media History Lesson

Analog Mass Media: Radio, TV & Video Recording

and Preparation for next Lectures

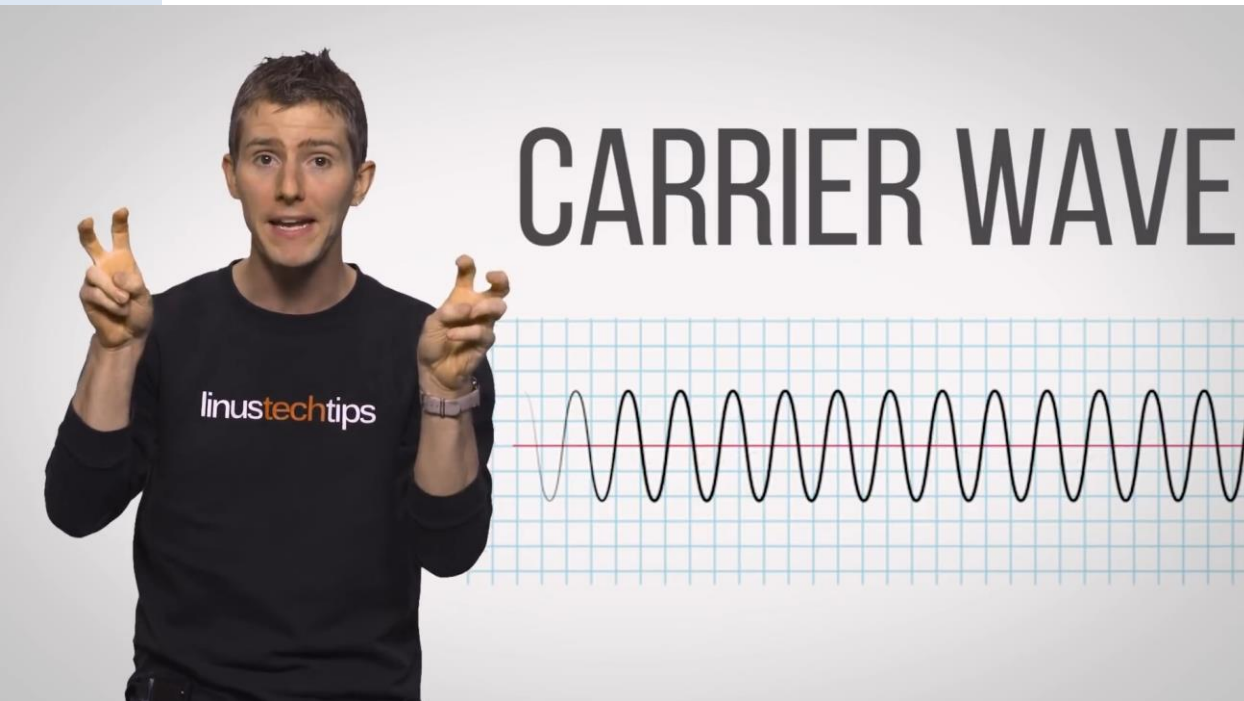
HDR Video, Modern Codecs and Filmmaking

Important Questions

- What is modulation?
- What is a carrier wave?
- What is the purpose of modulation?
- What are the 2 different kinds of modulation that are used for AM and FM radio?
- What is the basic functionality/components needed in a radio?

How AM and FM (=UKW) Radio works

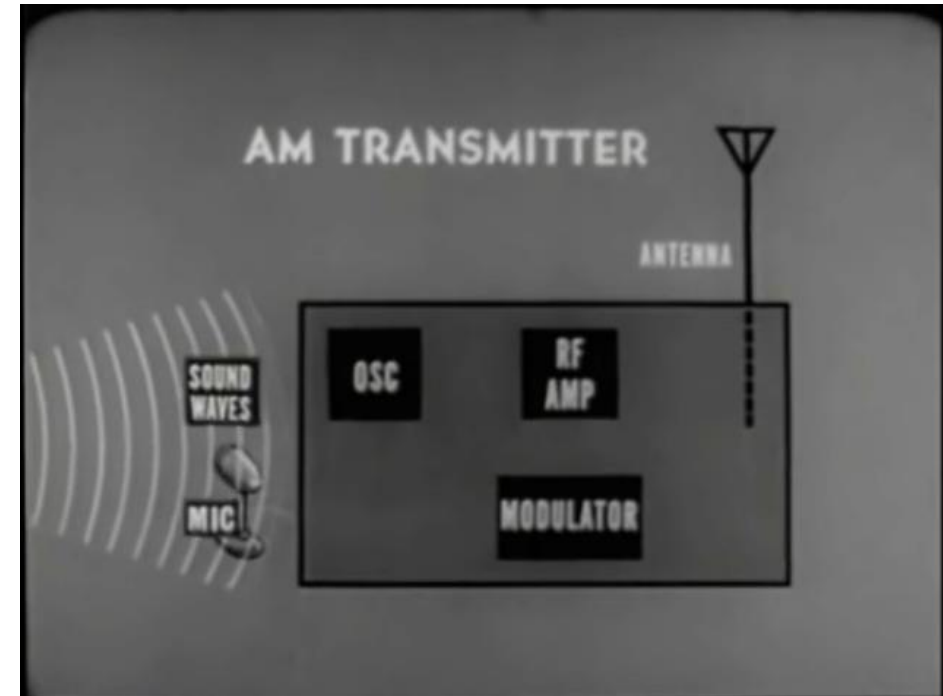
Simple fast explanation



<https://www.youtube.com/watch?v=w3M4cqAMWQs>

Very good, more detailed explanation

Highly edited version of US Army training video (TF11-3482 - Frequency Modulation Part I: Basic Principles, 1964) on the basics of AM & FM transmission.

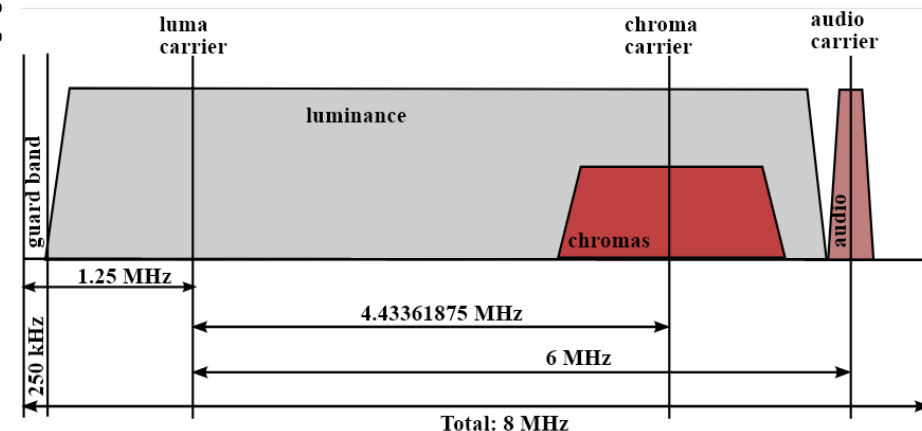


Hint: Turn on
Captions
(Subtitles) in
Youtube for
better
understanding

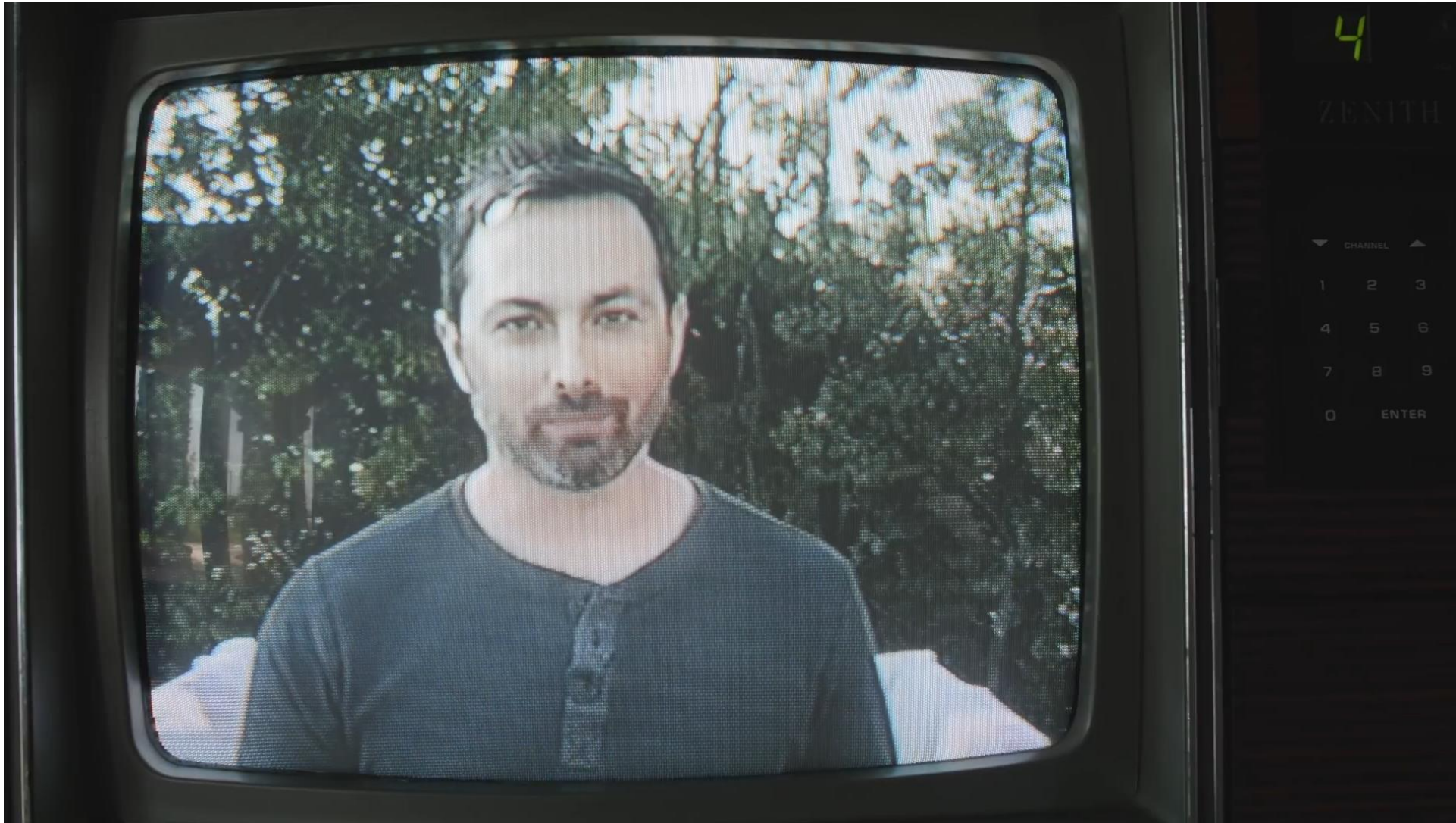
<https://www.youtube.com/watch?v=D65KXwfDs3s>

Relevant Questions

- How does a cathode ray tube TV work?
- How are colored dots on an analog TV generated?
- What is interlacing?
- How does a vacuum tube camera based on an electron beam work?
- In addition: Look up the PAL standard:
 - How many visible lines are displayed?
 - Find out what this diagram shows:



How Was Video Invented?



Relevant Questions

- Explain at least 2 methods that were used to store data before hard disk drives were invented.
- What are examples of portable storage before the invention of solid state storage like USB Sticks?
- How is data stored on an optical disc (e.g. CD / DVD) ?
- Look up online: How much faster is internal RAM compared to SSD storage?

Brief History of (Digital) Storage



In Preparation for next Lecture

- Overview of Codecs & Streaming
- Explanation of the 3 relevant HDR Video Standards & Differences

Relevant Questions

- What is the main difference between HDR 10 and Dolby Vision?
- What kind of metadata is transmitted in Dolby Vision?
- What is Adaptive Bitrate Streaming and which standards are mentioned that use it?

HDR Video Standards Explained



<https://www.youtube.com/watch?v=tzm2XjcyKDQ>

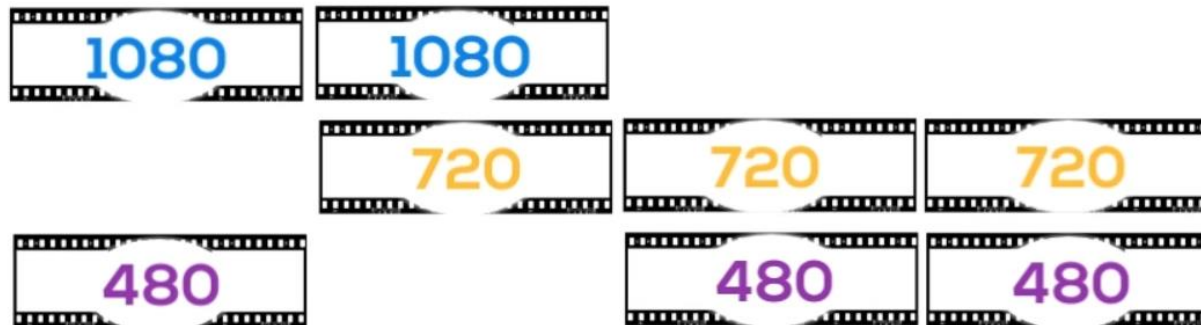
Video Codecs Containers Today

VIDEO FORMATS

FORMATS

MP4

Adaptive Bitrate Segmentation



MEDIUM

SLOW

FAST

FAST



Relevant Questions

- If you do not understand the video on the next slide, watch this one first

“Explaining Digital Video: Formats, Codecs & Containers”

<https://www.youtube.com/watch?v=-4NXxY4maYc>

(compression will be explained within the lecture)

What you should know after watching the video(s):

- What is a video container? (explained in the video above)
- What are common video codecs? (explained in the video above)
- What is the Profile of a Codec? What is contained in the profile?
- What are the Levels of a Codec? Give an example of what is specified in a level.
- For which purposes is the H.264 codec used?
- What are disadvantages of H.264 compared to newer codecs?
- What is “All-I” (All-Intra) video compression? (should become more clear in our next lecture)
- What is the purpose of an Intermediary Codec?
- Which two Intermediary Codecs are mentioned and what is the advantage of these intermediary codecs?
- What is the disadvantage of using ProRes and DNxHR for normal video recording?

Modern Video Codecs & Workflow



Hint: Reduce playback rate to **at least** a factor of 0.9 or 0.85

Relevant Questions

- What is the purpose of an LED wall or green screen?
- What are the advantages of using LED walls over green screens?
- Why is it necessary to motion track cameras in a virtual set?
- What technology is used and needed in a virtual set consisting of LED screens?
- How is it possible to change the virtual set in real time?
- Optional:
 - If interested, see also the optional video on the last slide about color grading in film with LUTs
 - There is an optional video about previsualization in movies (see below) on the example of recent Marvel movies

Modern Film Making: Virtual Set instead of Green Screen



Optional Videos

Relevant Questions

- What is the challenge building a tape video recorder?
- What is the innovation to solve these challenges?
- How is video and audio information stored on a video tape?

- Optional - if you are interested:

How cassette tape recorders work

<https://electronics.howstuffworks.com/gadgets/audio-music/cassette.htm>

Optional: Video Cassette Recording Explained



<https://www.youtube.com/watch?v=KfuARMCyTvg>

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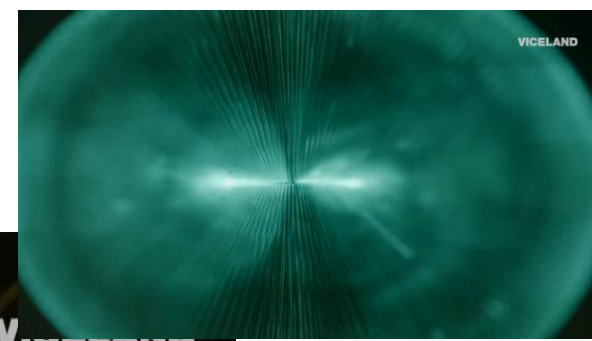
Relevant parts (in minutes):

Beginning → 7:00 : Development of a consumer VCR

9:30 – 10:00 : How the signal is stored on the tape

11:40 - end : Demonstration of VHS quality & SVHS explained

Optional: Scanimate



Hint: Turn on
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understanding.

Relevant Questions

- What are 2 fields in interlaced video?
- At position 5:45min in the video: Explain how quantization is related to this statement.
 - What are disadvantages of analog signal transmission or storage?
- What is the advantage of progressive scan?
- What is the factor of pixels when scaling from Full HD to 4K or from 4K to 8K resolution?

Optional: Home Video: Crash Course Film History



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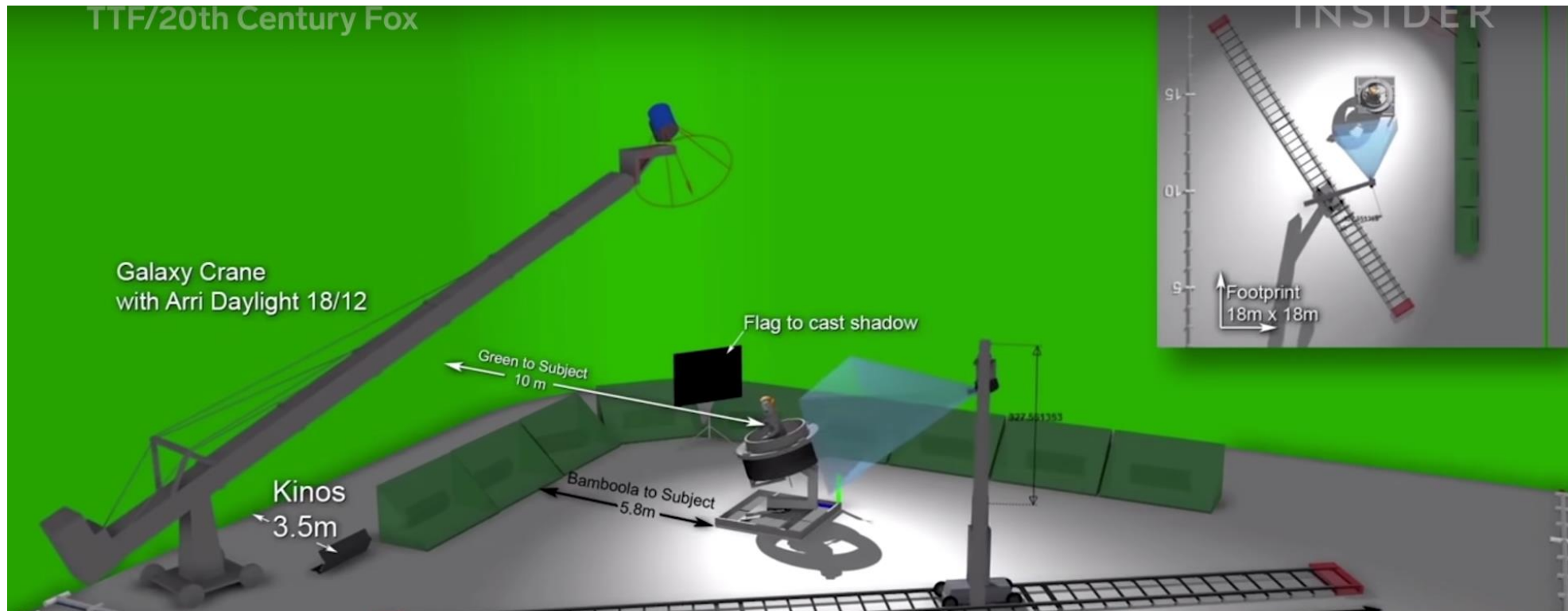
<https://www.youtube.com/watch?v=VfJ-6nQAmTk>

Optional: Excellent Video about Dynamic Range – From a Filmmaker



<https://www.youtube.com/watch?v=2sshGdMgJxQ>

Modern Film Making – The Role of Previsualization (in Marvel Movies)



<https://www.youtube.com/watch?v=bgvvgi3ShcmY>

Optional

Color Grading Explained: LUTs (Look Up Tables)

- Read through this website
<https://www.studiobinder.com/blog/what-is-lut/>
- The first video “LUTS: The complete video guide” gives a good introduction (expecially from 3:00 – 5:30min)
- Understand 1D LUTs and 3D LUTs